Nonlinear 3D Euler: checked solver foundations

Active research goal • draft PR #90 • progress record, 2 November 2026

Accuracy and optimality remain unproved. Lean has checked the named foundations listed below. The result is not yet a solver certified to reach a specified physical error, and no matching total-work lower bound has been established.

The physical problem

Full, unforced, compressible Euler with $\gamma = 2$ on the periodic cube $[0, 2\pi]^3$. The family has amplitude $A \in [0.5, 1]$ and density perturbation $\delta \in [-0.05, 0.05]$. Only initial conditions are supplied; all five conserved fields must evolve.

$\rho_0 = 1 + \delta \sin x \sin y \sin z$
$v_0 = A (\sin x \cos y \cos z, -\cos x \sin y \cos z, 0)$
$p_0 = 1 + (A^2/16)(\cos 2x + \cos 2y)(\cos 2z + 2)$
$E_0 = p_0 + \rho_0 |v_0|^2/2$

Initially $\rho_0 \geq 0.95$ and $p_0 \geq 0.625$. At $(0, 0, \pi/4)$, the vertical momentum equation forces $\partial_t(\rho v_z) = A^2/4 \geq 1/16$. This rules out freezing the initial zero vertical velocity. It does not prove that no analytic representation could ever exist.

Candidate and intended result

The current candidate is a conservative, explicit, global-speed Rusanov finite-volume method with exact-formula initial cell averages and a final-time-clipped CFL step. $T = 0.25$ is the numerical target; a common smooth existence interval is still unproved. The intended error includes the full evolved field, spatial reconstruction and time integration, using volume-normalized L1 error in each conserved component.

Required for completion	Current status
Common smooth time and uniform regularity	Unproved
Fully discrete nonlinear stability and continuum error	Unproved; foundations checked
Scientifically justified competitor class	Initial search scope; not finalized
Matching total-work upper and lower bounds	Unproved

All competitors receive the explicit initial-condition formulas. The target is not initial-data sampling complexity, a cache schedule, a CFL count, an output-size bound, or a ranking of a finite benchmark catalogue.

What Lean has checked

The retained audit checks **21 named statements** with Lean (version 4.24.0, x86_64-unknown-linux-gnu, commit 797c613eb9b6d4ec95db23e3e00af9ac6657f24b, Release). Transitive axiom dependencies are limited to the standard principles propext, Classical.choice and Quot.sound. Statements must still be interpreted with their definitions and explicit premises.

Layer	Checked claim and its limit
Actual nonlinear finite-volume update	Exact-real positive density/pressure from computed wave speeds and a valid CFL ratio; periodic conservation of all five components. This does not prove convergence.
Conservative / primitive connection	Actual derivative rules, invertible state-map derivative, equivalent Euler residuals, and a positive symmetrizing energy. This does not construct a solution.
Frozen principal update	The constant-coefficient linearized update contracts its discrete energy under the stated wave-speed and CFL bounds.
Varying principal update	An exact energy defect and an explicit growth estimate from stated spatial difference bounds. This is not the full nonlinear update stability theorem.
Grid embedding and state-box bounds	A mesh-independent 3D embedding and explicit coercivity/coefficient bounds. The trajectory's derivative energies and state-box membership are not bounded yet.

The new stability estimate

Write $q = 1/\rho$, $e = (e_q, e_v, e_p)$ and $g_{q,p}(e,e) = e_p^2/(2p) + |e_v|^2/q + (e_q/q + e_p/(2p))^2$. Sum this over cells to obtain E. For the principal update, the proof gives
$\mathbf{E_{after} \le [1 + 756\,\Delta t\,(256\,B\,K + 32\,D)]\,E_{before}}$.

Both energies use the old local metric. Premises include $1/4 \le q,p \le 4$, $|a_d| + |v_d| \le B$, a valid CFL ratio, an identity central shift, and neighbor differences satisfying $|\Delta q| + |\Delta p| \le hK$ and $|\Delta v_d| \le hD$. Here Δt = ratio × h. The constant does not depend on the number of cells when B, K and D have uniform bounds. Proving such bounds, the time variation of the metric and the nonlinear remainder are outstanding steps.

Exact negative control: with two cells, $q = (1,2)$, $p = 1$, $v = 0$, directional speeds 2 and ratio 1/12, the CFL number is 1/2. A specific error field has energy 5 before and 125/16 = 7.8125 after the principal update. Thus a varying local metric need not contract. This error-field example does not replace the nonlinear Taylor–Green research family.

Grid control and numerical evidence

The checked embedding gives $|f(i,j,k)|^2 \le 8\, E_3(f)$. In one dimension, $E_1(f) = n^{-1} \Sigma f^2 + n \Sigma(f_{k+1} - f_k)^2$; E_2 and E_3 apply this construction recursively in the other directions. It contains the eight mixed differences with each coordinate differentiated at most once, scaled to the unit cube. The constant 8 is independent of n. This controls values from derivative energies; it does not bound those energies along the Euler evolution.

Independent candidate checks

Initial cell averages agree with independent eight-point-per-axis Gauss quadrature to 9.99e-16 in the tested cells. A randomized admissible-field update has conservation drift 3.55e-15. An intentionally too-small split speed gives pressure −49, independently reproduced by Lean. These checks do not certify floating-point error or continuum accuracy.

n	A / δ	Steps	min ρ	min p	max |v_z|
16	0.5 / -0.05	11	0.964880	0.971706	0.008675
32	0.5 / -0.05	23	0.957623	0.944254	0.012397
16	0.75 / 0	12	0.997845	0.944951	0.018992
32	0.75 / 0	25	0.997943	0.882108	0.027463
16	1 / 0.05	13	0.969251	0.916210	0.032740
32	1 / 0.05	27	0.961357	0.803544	0.047906

All six runs end at T = 0.25. They show evolving vertical velocity and admissible sampled states on two grids. Their success neither proves the full parameter family remains smooth nor supplies an exact continuum reference. No speedup or optimality claim is inferred from these runs.

Reproducibility and next proof obligations

The source is under demos/lean_euler/nonlinear/formal. record/solver contains the frozen assembled Lean input, generated audit, logs, result JSON, component hashes, numerical checks and local-lab job receipts. The Python method is not a verified extraction of the Lean definitions.

Run from the repository root with a new output directory:
.venv/bin/python demos/lean_euler/nonlinear/run_solver_study.py
--output artifacts/nonlinear-energy-replay

The next required bridge is quantitative regularity and a stability/error estimate for the actual nonlinear update. Initialization and continuous reconstruction must then be connected to the evolved PDE solution. A complete cost model and a matching necessary-accuracy lower bound remain essential before the goal can be completed.

Assembled proof SHA-256: 769f19ff16454574b600707bc1fceafd
b261ec6971d0195e0304122c9994451f
Draft PR: github.com/tomzhu0225/simjecture/pull/90